Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Electric Dipole and Energy Density

Example 1

Two dipoles with dipole moments $-5\mathbf{a}_z$ nC · m and $9\mathbf{a}_z$ nC · m are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.

Solution:

$$\begin{aligned} V &= \sum_{k=1}^{2} \frac{\mathbf{p}_k \cdot \mathbf{r}_k}{4\pi\varepsilon_o r_k^3} \\ &= \frac{1}{4\pi\varepsilon_o}\left[\frac{\mathbf{p}_1 \cdot \mathbf{r}_1}{r_1^3} + \frac{\mathbf{p}_2 \cdot \mathbf{r}_2}{r_2^3}\right] \end{aligned}$$

where

$$\begin{aligned} &\mathbf{p}_1 = -5\mathbf{a}_z, \quad \mathbf{r}_1 = (0, 0, 0) - (0, 0, -2) = 2\mathbf{a}_z, \quad r_1 = |\mathbf{r}_1| = 2 \\ &\mathbf{p}_2 = 9\mathbf{a}_z, \quad \mathbf{r}_2 = (0, 0, 0) - (0, 0, 3) = -3\mathbf{a}_z, \quad r_2 = |\mathbf{r}_2| = 3 \end{aligned}$$

Hence,

$$\begin{aligned} V &= \frac{1}{4\pi \cdot \dfrac{10^{-9}}{36\pi}}\left[\frac{-10}{2^3} - \frac{27}{3^3}\right] \cdot 10^{-9} \\ &= -20.25 \text{ V} \end{aligned}$$

> **PRACTICE EXERCISE 4.13**
>
> An electric dipole of $100\,\mathbf{a}_z$ pC · m is located at the origin. Find V and $\mathbf{E}$ at points
>
> (a) $(0, 0, 10)$
> (b) $(1, \pi/3, \pi/2)$
>
> **Answer:** (a) 9 mV, $1.8\mathbf{a}_r$ mV/m, (b) 0.45 V, $0.9\mathbf{a}_r + 0.7794\mathbf{a}_\theta$ V/m.

Example 2

The point charges -1 nC, 4 nC, and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, and $(1, 0, 0)$, respectively. Find the energy in the system.

Solution:

Method 1:

$$
\begin{aligned}
W &= W_1 + W_2 + W_3 \\
&= 0 + Q_2 V_{21} + Q_3(V_{31} + V_{32}) \\
&= Q_2 \cdot \frac{Q_1}{4\pi\varepsilon_o \,|(0,0,1) - (0,0,0)|} \\
&\quad + \frac{Q_3}{4\pi\varepsilon_o}\left[\frac{Q_1}{|(1,0,0) - (0,0,0)|} + \frac{Q_2}{|(1,0,0) - (0,0,1)|}\right] \\
&= \frac{1}{4\pi\varepsilon_o}\left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}}\right) \\
&= \frac{1}{4\pi \cdot \dfrac{10^{-9}}{36\pi}}\left(-4 - 3 + \frac{12}{\sqrt{2}}\right) \cdot 10^{-18} \\
&= 9\left(\frac{12}{\sqrt{2}} - 7\right) \text{nJ} = 13.37 \text{ nJ}
\end{aligned}
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